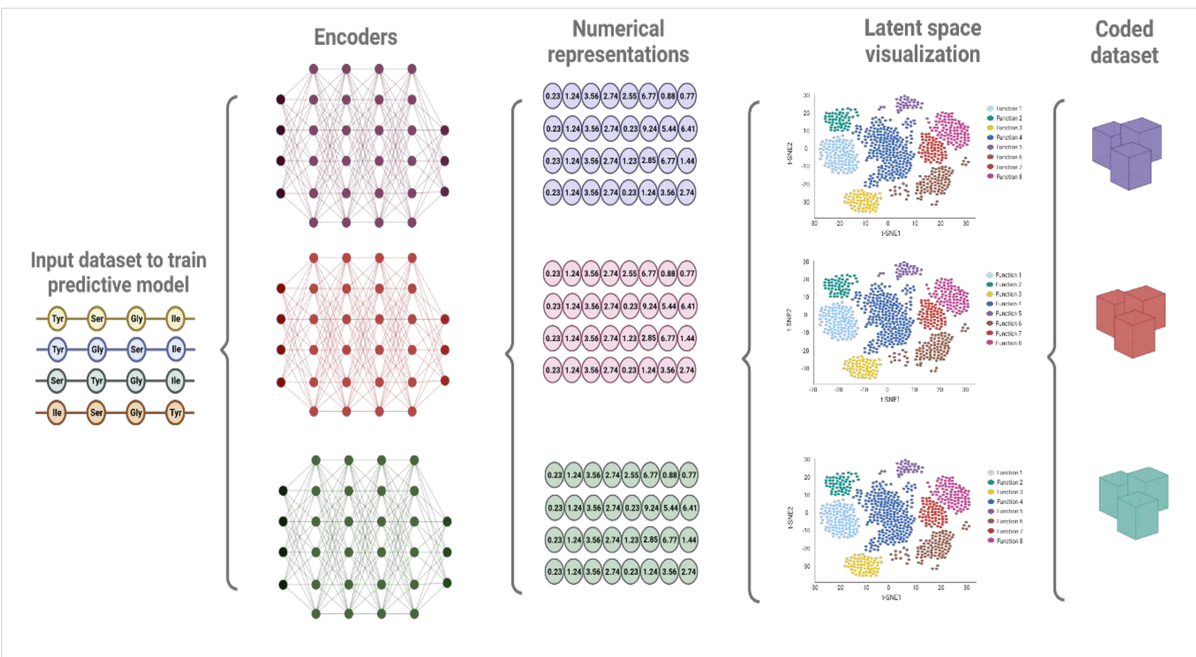
Exploring machine learning algorithms and protein language model strategies to develop functional enzyme classification systems

Diego Fernández[1], Álvaro Olivera-Nappa[2], Roberto Uribe-Paredes[1], David Medina Ortiz[1,2]
[1]Departamento de Ingeniería en Computación, Universidad de Magallanes, [2]Centre for Biotechnology and Bioengineering, Universidad de Chile
Contact: david.medina@umag.cl or david.medina@cebib.cl

Motivation and objectives

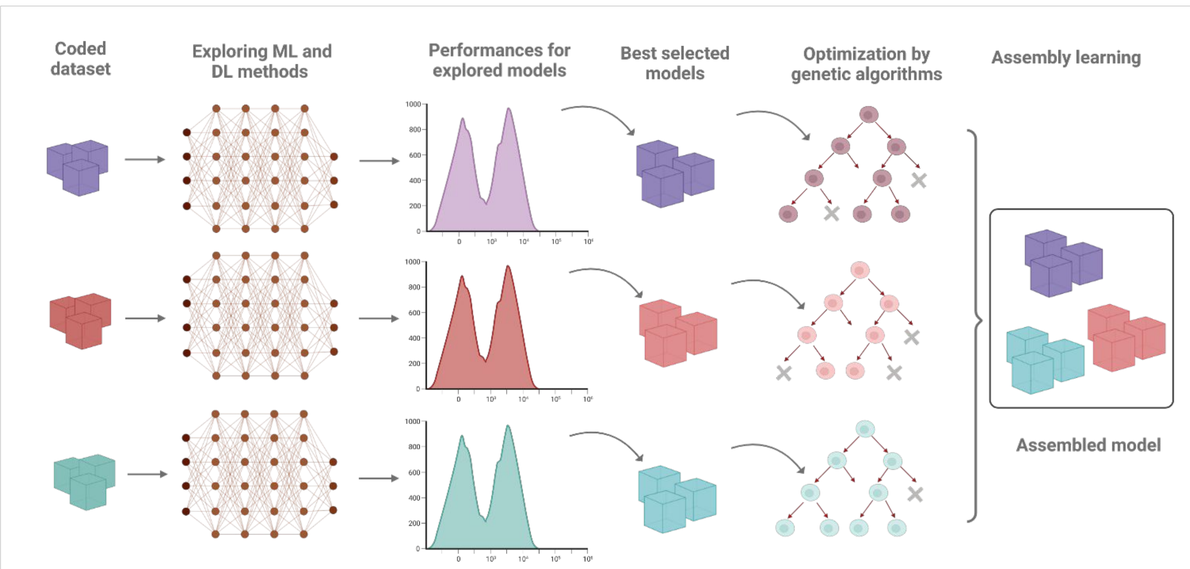
Discovering functionalities for unknown enzymes has been one of the most common bioinformatics tasks. Functional annotation methods based on phylogenetic properties have been the gold standard in every genome annotation process. However, these methods only succeed if the minimum requirements for expressing similarity or homology are met. Alternatively, machine learning and deep learning methods have proven helpful in this problem, developing functional classification systems in various bioinformatics tasks. Nevertheless, there needs to be a clear strategy for elaborating predictive models and how amino acid sequences should be represented.

Using protein language models as numerical representation strategies



Different numerical representation strategies were explored for protein sequences. Amino acid encoding techniques were applied using physicochemical properties extracted from the AAIndex database. Besides, the Fast Fourier transform was employed to represent the amino acid coding results as frequency signals. In both cases, zero-padding strategies were utilized to ensure equal vector size. Finally, protein language models were applied to represent numerically the enzyme sequences. This work used different pre-trained models available in the literature, such as Bert, bepler, ESM1b, and Prottrans.

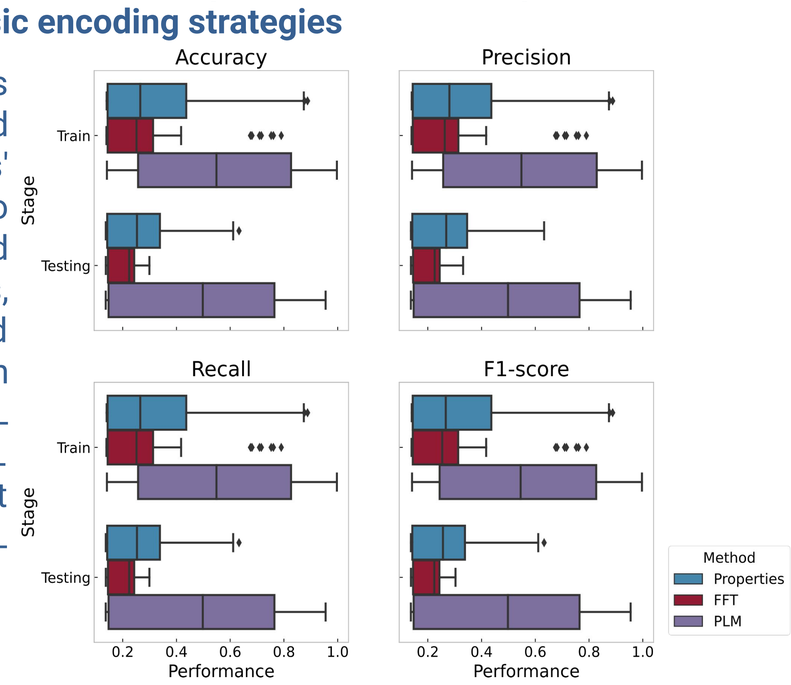
A generalizable pipeline to build functional classification systems



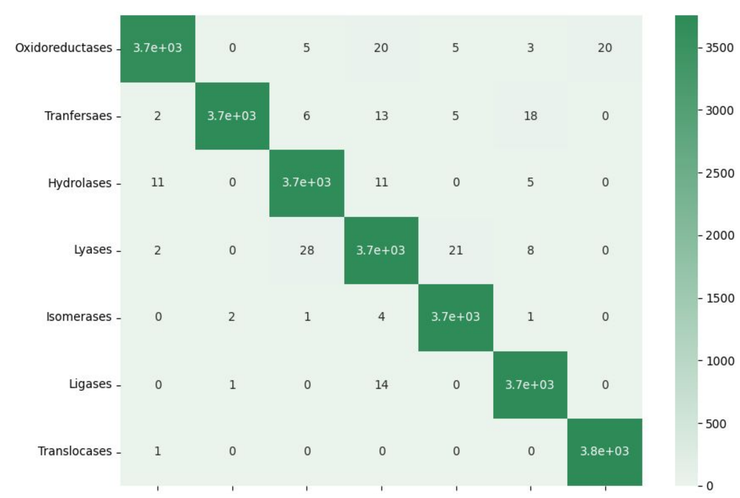
The training of functional classification models was based on applying machine learning algorithms and deep learning architectures to explore predictive modeling strategies, evaluated via classic performance metrics such as the precision, recall, f-score, and MCC coefficient. Then, statistical methods were used to select the best combinations of algorithms and numerical representations strategies, whose hyperparameters were optimized in a tuning process via genetic algorithms. Finally, ensemble learning strategies combined the hyperparameter-optimized models into a single predictive system were applied.

Protein language models achieve better performance in comparison to classic encoding strategies

Numerical representations supported by NLP improved classification models' performances compared to classical amino acid encoding methods. Besides, CNN architectures achieved better performance in comparison with classic ML algorithms. However, the DL architectures tend to present overfitting compared to ML methods.

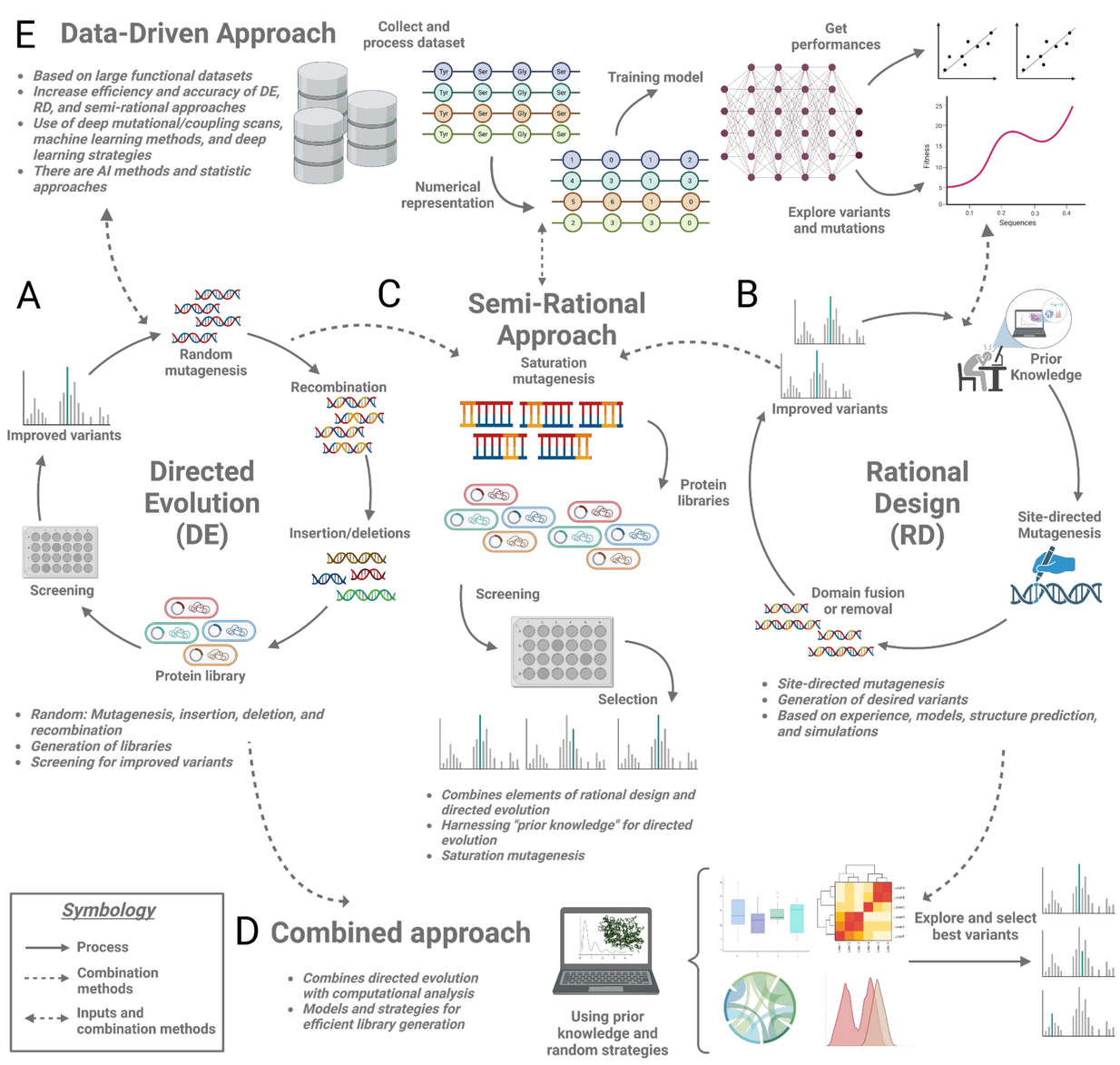


The assembled model improves the performance of the individual classification models



The assembled model comprises ten optimized methods that combine NLP strategies, ML methods, and CNN architectures. The assembled model achieves a 98% precision in training and testing steps and a 0.89 MCC, demonstrating the usability of the ensemble learning process.

Future work: Toward an autonomous design of enzyme guide by machine learning strategies



Funding

The authors acknowledge funding by the MAG-2095 project, Ministry of Education, Chile. DMO acknowledges ANID for the project "SUBVENCIÓN A INSTALACIÓN EN LA ACADEMIA CONVOCATORIA AÑO 2022", Folio 85220004. DMO and AON gratefully acknowledge support from the Centre for Biotechnology and Bioengineering - CeBiB (PIA project FB0001, Conicyt, Chile). RUP acknowledges ANID for Fondecyt project 1230298.